ITDS 124 Mathematics of Computing

Chapter 2.3
Part 2: Introduction to Proof

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Note on this lecture

- This lecture has been developed from several textbooks including
  - Discrete Mathematical Structures by Bernard Kolman, Robert Busby, et al.
  - Discrete Mathematics and Its Applications by Kenneth Rosen
  - Discrete Mathematics: An Open Introduction by Oscar Levin

Some terminology before going to proofs (1)

- A **theorem** is a statement that can be shown to be true.
  - In mathematical writing, the term theorem is usually reserved for a statement that is considered at least somewhat important.
- **Propositions** are sometimes used to call less important theorems.
- **Axioms** are statements we assume to be true.
- A **lemma** is a less important theorem that is helpful in the proof of other results.
  - Complicated proofs are usually easier to understand when they are proved using a series of lemmas, where each lemma is proved individually.

Some terminology before going to proofs (2)

- A **corollary** is a theorem that can be established directly from a theorem that has been proved.
- A **conjecture** is a statement that is being proposed to be a true statement, usually on the basis of some partial evidence, a heuristic argument, or the intuition of an expert.
  - When a proof of a conjecture is found, the conjecture becomes a theorem.
  - Many times, conjectures are shown to be false, so they are not theorems.

Introduction to Proofs

- A **proof** is a valid argument that establishes the truth of a theorem.
- Ingredients of a proof can be
  - hypotheses/premises of the theorem,
  - axioms assumed to be true, and
  - previously proven theorems.
- Using these ingredients and rules of inference, the final step of the proof establishes the truth of the statement being proved.

Formal proofs vs. Informal proofs

- The arguments to show that statements involving propositions and quantified statements are true are called **formal proofs**, where all steps were supplied, and the rules for each step in the argument were given.
  - Formal proofs of useful theorems can be extremely long and hard to follow.
  - Computers are better at producing formal proofs using automated reasoning systems.

- **Informal proofs** are used to explain to humans why theorems are true, where more than one rule of inference may be used in each step, where steps may be skipped, where the axioms being assumed and the rules of inference used are not explicitly stated.

Proving mathematical theorems can be difficult

- Anyone who doesn't believe there is creativity in mathematics clearly has not tried to write proofs.
- Finding a way to convince the world that a particular statement is necessarily true is a mighty undertaking and can often be quite challenging.

- **There is not a guaranteed path to success in the search for proofs.**
- For example, in the summer of 1742, a German mathematician by the name of Christian Goldbach wondered whether every even integer greater than 2 could be written as the sum of two primes. Centuries later, we still don't have a proof of this apparent fact (computers have checked that "Goldbach's Conjecture" holds for all numbers less than 4×10^{18}, which leaves only infinitely many more numbers to check).

Writing proofs

- Writing proofs is a bit of an art. Like any art, to be truly great at it, you need some sort of inspiration, as well as some foundational technique.
- A good place to start is to study existing proof methods/techniques.
- These methods provide the overall approach and strategy of proofs.
- Understanding these methods is a key component of learning how to read and construct mathematical proofs.
- Once we have chosen a proof method, we use axioms, definitions of terms, previously proved results, and rules of inference to complete the proof.

Forms of Theorems

- Many theorems assert that a property holds for all elements in a domain, such as the integers or the real numbers.
- Often the universal quantifier (needed for a precise statement of a theorem) is omitted by standard mathematical convention.

For example, the statement:

" If $x > y$, where x and y are positive real numbers, then $x^2 > y^2$. "

really means

" For all positive real numbers x and y, if $x > y$, then $x^2 > y^2$."

Proving Theorems

- Many theorems have the form:

$$\forall x\,(P(x) \rightarrow Q(x))$$

- To prove them, we show that where c is an arbitrary element of the domain, $P(c) \rightarrow Q(c)$
- By universal generalization the truth of the original formula follows.
- So, we must prove something of the form: $p \rightarrow q$

A direct proof of a conditional statement $p \rightarrow q$

- You will find that direct proofs of many results are quite straightforward, with a fairly obvious sequence of steps leading from the hypothesis to the conclusion.
- In a direct proof, we assume that p is true and use axioms, definitions, and previously proven theorems, together with rules of inference, to show that q must also be true.
- The general format to prove $p \rightarrow q$ is this:

Assume p. Explain, explain, . . . , explain. Therefore q.

Even and Odd Integers

Definition:

- The integer n is even if there exists an integer k such that $n = 2k$.
- The integer n is odd if there exists an integer k, such that $n = 2k + 1$.
- Note that every integer is either even or odd and no integer is both even and odd.

- We will need this basic fact about the integers in some of the example proofs to follow.

Direct proof : Example 1

Give a direct proof of the theorem "If n is an odd integer, then n^2 is odd."

Solution:

Assume that _____.

Then $n =$ _____ for an integer k. Squaring both sides of the equation, we get:
$n^2 =$ _____ $=$ _____ $=$ _____ $=$ _____,
where _____, an integer.

By definition of an odd integer, we can conclude that _____.

Consequently, we have proved that if n is an odd integer, then n^2 is an odd integer.

Rational number

Definition:

- The real number r is *rational* if there exist integers p and q where $q \neq 0$ such that $r = \frac{p}{q}$

Direct proof : Example 2

Prove that the sum of two rational numbers is rational.

Solution*:*

Indirect proofs

- Direct proofs lead from the premises of a theorem to the conclusion.
  - They begin with the premises, continue with a sequence of deductions, and end with the conclusion.
- However, many attempts at direct proofs often reach dead ends.
- We need other methods of proving theorems of the form $\forall x\,(P(x) \rightarrow Q(x))$.
- **Indirect proofs** of theorems of this type do not start with the premises and end with the conclusion.

Proof by Contraposition

- Proofs by contraposition make use of the fact that the conditional statement $p \to q$ is **logically equivalent** to its **contrapositive**, $\sim q \to \sim p$.
- This means that the conditional statement $p \to q$ can be proved by showing that its contrapositive, $\sim q \to \sim p$, is true.
- There are plenty of examples of statements which are hard to prove directly, but whose contrapositive can easily be proved directly.
- In a proof by contraposition of $p \to q$, we take $\sim q$ as a premise, and using axioms, definitions, and previously proven theorems, together with rules of inference, we show that $\sim p$ must follow.
- The skeleton of the proof of $p \to q$ by contrapositive will look roughly like this:

Assume $\sim q$. Explain, explain, . . . explain. Therefore $\sim p$.

Proof by Contraposition : Example 1

Prove that if n is an integer and $3n + 2$ is odd, then n is odd.

Solution 1: We <u>first attempt a direct proof</u>.

Assume that $3n + 2$ is an odd integer.

This means that $3n + 2 = 2k + 1$ for some integer k.

Can we use this fact to show that n is odd? We see that $3n + 1 = 2k$, but there does not seem to be any direct way to conclude that n is odd.

So our attempt at a direct proof failed.

Proof by Contraposition : Example 1 (Continue)

Prove that if n is an integer and $3n + 2$ is odd, then n is odd.

Solution 2: We next try a proof by contraposition.

Assume _____. So, _____for some integer k.

Thus $3n + 2 =$ _____ $=$ _____ $=$ _____ $=$ _____ for _____.

Therefore _____.

Since we have shown $\sim q \rightarrow \sim p$, so $p \rightarrow q$ must hold as well. We have proved the theorem "If n is an integer and $3n + 2$ is odd, then n is odd.

Proof by Contraposition : Example 2

Prove that for an integer n, if n^2 is odd, then n is odd.

Solution: Use proof by contraposition.

Proofs of statements which are not in the implication form

- So far, we have learned how to write direct proofs and indirect proofs of conditional statements $p \to q$.

- There might be statements which really cannot be rephrased as implications.

- For example, prove that "$\sqrt{2}$ is irrational."

- In this case, it is hard to know where to start.

Proofs by Contradiction

- Suppose we want to prove that a statement p is true.
- Also, suppose that we can find a contradiction q such that $\sim p \rightarrow q$ is true.
- Because q is false, but $\sim p \rightarrow q$ is true, we can conclude that $\sim p$ is false, which means that p is true.

- How can we find a contradiction q that helps us prove that p is true in this way?
- Because the statement $r \wedge \sim r$ is a contradiction whenever r is a proposition, we can prove that p is true if we can show that $\sim p \rightarrow (r \wedge \sim r)$ is true for some proposition r.

Assume $\sim p$ is true

try to show that $r \wedge \sim r$

- Proofs of this type are called **proofs by contradiction**.
- Because a proof by contradiction does not prove a result directly, it is another type of indirect proof.

Proof by Contradiction : Example 1

Fill in the details of the proof to create a contradiction between 2 statements

- Show that at least four of any 22 days must fall on the same day of the week.

Solution:

Let p be the proposition "At least four of 22 chosen days fall on the same day of the week."

Suppose that $\sim p$ is true. This means that at most three of the 22 days fall on the same day of the week.

Consequently, we know that p is true. We have proved that at least four of 22 chosen days fall on the same day of the week.

Proof by Contradiction : Example 1

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Let p be the proposition "At least four of 22 chosen days fall on the same day of the week."

Suppose that $\sim p$ is true. This means that at most three of the 22 days fall on the same day of the week.

Because there are seven days of the week, this implies that at most 21 days could have been chosen, as for each of the days of the week, at most three of the chosen days could fall on that day. This contradicts the premise that we have 22 days under consideration.

That is, if r is the statement that 22 days are chosen, then we have shown that $\sim p \rightarrow (r \wedge \sim r)$.

Consequently, we know that p is true. We have proved that at least four of 22 chosen days fall on the same day of the week.

Proof by Contradiction : Example 2

- Prove that $\sqrt{2}$ is irrational.
- Solution.

  Let p be the statement $\sqrt{2}$ is irrational.

  Suppose that _____ is true. Then $\sqrt{2}$ is _____ and equal to _____.

  Thus $\sqrt{2}$ is irrational.

Fill in the details of the proof to create a contradiction between 2 statements

Proofs of biconditional statement $p \leftrightarrow q$

- To prove a theorem that is a biconditional statement $p \leftrightarrow q$, we show that $p \rightarrow q$ and $q \rightarrow p$ are both true.
- The validity of this approach is based on the tautology

$$(p \leftrightarrow q) \leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p).$$

Proofs of biconditional statement: Example 1

- Prove the theorem "If n is an integer, then n is odd if and only if n^2 is odd."

Solution:

Proofs by Counterexample

- To show a statement of the form $\forall x\ P(x)$ is false, we need only find a counterexample, that is, an example x for which $P(x)$ is false.
- There might be more than one counterexamples. Showing one is enough.

Proofs by Counterexample: Example 1

- Show that the statement "Every positive integer is the sum of the squares of two integers" is false.

Solution:

To show that the statement is false, we look for a counterexample, which is a particular integer that is not the sum of the squares of two integers.

Because 3 cannot be written as the sum of the squares of two integers. To show this is the case, note that the only perfect squares not exceeding 3 are $0^2 = 0$ and $1^2 = 1$.

Furthermore, there is no way to get 3 as the sum of two terms each of which is 0 or 1.

Consequently, we have shown that "Every positive integer is the sum of the squares of two integers" is false.

Proof by Cases

- Sometimes we cannot prove a theorem using a single argument that holds for all possible cases. We need a method that can be used to prove a theorem, by considering different cases separately.

- To prove a conditional statement of the form $(p_1 \vee p_2 \vee \cdots \vee p_n) \rightarrow q$ the tautolog $[(p_1 \vee p_2 \vee \cdots \vee p_n) \rightarrow q] \leftrightarrow [(p_1 \rightarrow q) \wedge (p_2 \rightarrow q) \wedge \cdots \wedge (p_n \rightarrow q)]$ can be used as a rule of inference.

- This shows that the original conditional statement with a hypothesis made up of a disjunction of the propositions $p_1, p_2, \ldots, p_n$ can be proved by proving each of the n conditional statements $p_i \rightarrow q, \quad i = 1,2, \ldots, n$, individually.

- Such an argument is called a **proof by cases**.

- Sometimes to prove that a conditional statement $p \rightarrow q$ is true, it is convenient to use a disjunction $p_1 \vee p_2 \vee \cdots \vee p_n$ instead of p as the hypothesis of the conditional statement, where p and $p_1 \vee p_2 \vee \cdots \vee p_n$ are equivalent.

Proof by Cases: Example 1

- Prove that if n is an integer, then $n^2 \geq n$ (or prove that for any integer n, $n^2 \geq n$).

Solution:

We can prove that $n^2 \geq n$ for every integer by considering three cases,

Case (i): When $n = 0$, we see that $0^2 \geq 0$.

Case (ii): When $n \geq 1$, when we multiply both sides by the positive integer n, we obtain $n \cdot n \geq n \cdot 1$. This implies that $n^2 \geq n$.

Case (iii): In this case $n \leq -1$. However, $n^2 \geq 0$. It follows that $n^2 \geq n$.

Because the inequality $n^2 \geq n$ holds in all three cases, we can conclude that if n is an integer, then $n^2 \geq n$.